Monetary Policy and Deposit Beta: The Role of Internal Capital Market *

Jing Ye†

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Abstract

This paper introduces a novel Internal Capital Market (ICM) channel—proxied by the parent debt ratio—to explain how Bank Holding Companies (BHCs) shape the transmission of monetary policy to their subsidiary banks. Using deposit beta as the primary measure of rate-setting behavior, I find that a higher parent debt ratio is significantly associated with lower deposit betas, indicating that the ICM prevents the full pass-through of the monetary policy rate to the deposit rate. Two distinct mechanisms underpin this relationship. First, a funding substitution effect, where internal capital transfers from the parent to its subsidiary banks reduce subsidiary banks' dependence on traditional deposits, thereby granting them greater flexibility in changing the deposit rate. Second, a risk-mitigation effect, in which higher parent debt ratio and cross-guarantee provisions incentivize the parent company to limit subsidiary bank's risk-taking, reducing the risk premiums demanded by depositors. Taken together, these findings suggest that parent company debt meaningfully weakens the sensitivity of deposit rates to monetary policy. Furthermore, parent debt can serve as an effective instrument for managing the capital adequacy of subsidiary banks.

Keywords: Deposit Beta, Bank Holding Company, Parent Debt, Subsidiary Bank, Capital Regulation, Monetary Policy

JEL classification: G21, G28, G32, G33, G38

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†Warwick Business School, University of Warwick. E-mail: jing.ye@warwick.ac.uk

1 Introduction

Monetary policy plays a crucial role in shaping banks' deposit rate. However, when the federal funds rate rises, deposit rates typically do not increase one-for-one and often remain below the policy rate. This sensitivity is measured by the *deposit beta*, which is estimated by regressing the change in a bank's interest expense on deposits (scaled by quarterly average assets) on contemporaneous and lagged changes in the federal funds rate, and then summing the coefficients (Drechsler et al. (2021)). Understanding the determinants of deposit beta is essential for assessing how effectively monetary policy is transmitted through the banking system, for understanding its impact on deposit beta, and for controlling the bank's deposit amount, which affects its lending activities.

Prior studies have examined how monetary policy affects banks' deposit rate without considering the ICM within BHCs. However, most large commercial banks in the U.S. operate as subsidiaries of BHCs, which have a natural ICM within them. This organizational structure raises an important question: what role does the ICM play in the transmission of monetary policy to subsidiary banks' deposit rate? Existing research typically focuses on the impact of monetary policy on deposit rates at the standalone or consolidated bank level without considering the impact of the ICM on subsidiary banks' deposit beta. Unlike standalone commercial banks, subsidiary banks within BHCs can access the ICM within their BHCs, where capital transfers from the parent company to subsidiary banks are possible. Therefore, this paper aims to fill the gap by focusing on the ICM within BHCs—proxied by the *parent debt ratio* (defined as the share of a BHC's non-deposit debt issued by its parent company)—and examining its effect on subsidiary banks' deposit beta.

The empirical results show that a higher parent debt ratio is associated with a lower deposit beta, suggesting that greater parent debt reduces subsidiary banks' sensitivity of deposit rates to the federal funds rate.

I examine two mechanisms, motivated by economic theory, regarding the function of the ICM within BHCs. First, I consider the funding substitution mechanism. The primary function of an ICM is the strategic movement of capital within the BHC. I investigate whether BHCs with higher parent debt ratios transfer more capital to their subsidiary banks. When a parent company injects capital into its subsidiary banks,

this capital serves as an alternative funding source, which reduces subsidiary banks' reliance on traditional deposits. This internal cushion grants subsidiary banks greater flexibility in managing their deposit rates and allows them to adjust deposit rates less aggressively during periods of monetary policy shifts. Consequently, I hypothesize that this mechanism leads to lower deposit betas and dampens sensitivity to both policy rate hikes and cuts. Second, I examine the risk-mitigation mechanism. This mechanism focuses on the disciplinary role of parent debt. When a parent company issues external debt to fund its subsidiary banks, it faces heightened incentives to curb subsidiary banks' risk-taking, largely due to cross-guarantee provisions (Ma et al. (2026)). As subsidiary bank risk decreases, depositors perceive greater safety and demand lower interest rate premiums (Martinez Peria and Schmukler (2001)). This reduction in risk-shifting behavior should, in theory, manifest as a lower deposit beta.

This paper highlights the function of internal capital markets within BHCs. Unlike existing studies—some of which focus on risk-sharing through internal capital markets among global banks (e.g., Cetorelli and Goldberg (2012))— I emphasize a distinct function of these markets: smoothing the transmission of monetary policy shocks to the deposit rates of domestic subsidiary banks within U.S. BHCs.

Although the empirical evidence shows a negative relationship between the parent debt ratio and deposit beta, it may be subject to common identification challenges. First, deposit beta may respond to deposit supply in addition to parent debt. For instance, if banks' lending opportunities decline as the Fed raises rates, then banks would make fewer loans and consequently take in fewer deposits by decreasing deposit beta even in the absence of the ICM channel. Furthermore, reverse causality may exist: a lower deposit beta may lead to greater issuance of parent debt. For example, when the subsidiary bank exhibits a low deposit beta, deposits are more likely to flow out as the federal funds rate increases. To maintain the same size of the subsidiary bank's balance sheet, the parent company may need to issue more external debt to provide the necessary funding for its subsidiary banks.

I address these identification challenges by exploiting the variation in an observable determinant of the parent debt ratio, the monetary policy shock constructed by Bauer and Swanson (2023), which provides plausibly exogenous variation in parent debt is-

suance. This approach allows me to identify the causal effect of the parent debt ratio on subsidiary banks' deposit beta. Using a two-stage least squares (2SLS) framework, I show that increases in the parent debt ratio correspond to significant declines in subsidiary banks' deposit beta, supporting the main inference.

I conduct several robustness checks to examine whether my main results are robust to alternative estimation of deposit beta and alternative proxies for deposit beta. Collectively, these tests help identify where the variation that drives the main results originates.

Although my findings provide new insights into the role of the ICM within BHCs in the transmission of monetary policy, they also raise important regulatory concerns regarding capital ratios at both the subsidiary bank and BHC levels. I illustrate this concern using a simple model in Section 6.

The remainder of this paper is organized as follows. Section 2 reviews the related literature. Section 3 develops the hypotheses through which parent debt reduces deposit beta. Section 4 describes the data sources, key variables, and the estimation of deposit beta. Section 5 presents the main empirical results, identification strategy, and robustness checks. Section 6 introduces a simple model, and Section 7 concludes.

2 Literature review

This paper contributes to several important strands of the monetary policy and banking literature. First, I introduce a new channel through which monetary policy transmits to banks' deposit rate. The existing literature on the deposit channel of monetary policy focuses on banks' deposit market power. Earlier work by Hannan and Berger (1997) and Neumark and Sharpe (1992) first empirically documented deposit price rigidity in markets with higher levels of concentration. Recent work by Drechsler et al. (2017) further shows that banks' deposit market power plays an important role in the transmission of monetary policy to deposit rates, finding that banks widen the deposit spreads they charge on deposits when the Fed funds rate rises, and deposits flow out of the banking system. Wang et al. (2022) examine the quantitative impact of bank market power on the transmission of monetary policy through the banking

system and finds that banks' market power plays an important role in determining the degree of monetary policy transmission and allows banks to charge higher markups on deposits, thus leading to a contraction in deposits and loans. Kho (2025) studies the transmission of monetary policy to deposit rates in the euro area with a focus on the role of banking sector concentration and asymmetric adjustment to monetary policy, finding that more concentrated domestic banking sectors pass on unexpected monetary tightening (easing) more slowly (quickly) than less concentrated banking sectors, suggesting that heterogeneity in the degree of banking sector concentration matters for the transmission of monetary policy to deposit rates. In this paper, I uncover the ICM channel within the broader deposit channel, where monetary policy influences banks' deposit rate sensitivity through its interaction with the ICM within the BHC, complementing the existing deposit channel.

Second, I connect the growing literature on the deposit channel of monetary policy to the ICM channel by emphasizing the role of the ICM in the transmission of monetary policy to deposit rates. The existing literature on the ICM within BHCs primarily focuses on the asset side of the entities' balance sheets without considering the liability side. Houston et al. (1997) examine the cash-flow sensitivity of loan growth at BHCs and finds that loan growth at subsidiary banks is more sensitive to the holding company's cash flow and capital position than to the bank's own cash flow and capital position. Furthermore, bank loan growth is negatively correlated with loan growth among other subsidiaries within the holding company. These findings suggest that BHCs establish the ICM to allocate scarce capital among their subsidiaries. Campello (2002) explores the ICM in financial conglomerates by comparing the responses of small bank subsidiaries and independent banks to monetary policy, finding that the ICM relax credit constraints faced by bank subsidiaries and lessen the impact of Fed policies on bank lending activity. Holod and Peek (2010) examine the ICM within multibank holding companies and finds that internal capital management within a multibank holding company involves not only the movement of capital to those subsidiaries with relatively greater need for capital but also the movement of assets (loans) from less well capitalized to better capitalized subsidiaries by means of loan sales and purchases among the subsidiaries. Cetorelli and Goldberg (2012) examines how the ICM of global banks

affect their liquidity management, response to liquidity shocks, and potential for international shock propagation. They find that global banks manage liquidity on a global scale by using cross-border internal funding in response to local shocks, and that having global operations insulates banks from changes in monetary policy. Consistent with Houston et al. (1997), my findings confirm the presence of the ICM within BHCs, which facilitate capital transfers from parent companies to their subsidiary banks. Moreover, complementing Campello (2002), I show that the ICM within BHCs mitigate the impact of Federal Reserve policy shocks on subsidiary banks' deposit beta. Unlike Cetorelli and Goldberg (2012), this paper focuses on domestic BHCs instead of global banks by examining the role of the ICM in the transmission of monetary policy to deposit rates.

Third, this paper relates to the literature about the low sensitivity of deposit rates to the federal funds rate by examining how internal capital transfers from parent companies to bank subsidiaries affect this low sensitivity. Earlier work by Neumark and Sharpe (1992) examine the impact of market concentration on the adjustment of prices in the market for consumer bank deposits and finds that banks in concentrated markets are slower to raise interest rates on deposits in response to rising market interest rates but faster to reduce them in response to declining market interest rates. Hannan and Berger (1997) examine the response of deposit rates to market rates and finds that deposit rate rigidity is greater in markets with high levels of concentration and that deposit rates are more rigid for increases than for decreases. Yankov (2022) examines the role of consumer search for the pricing of insured deposits and finds that high-search-cost depositors grant banks monopoly power and allow for low interest rate pass-through. Wang (2025) examines how the secular decline in interest rates affects banks' deposit spreads and credit supply, finding that a permanent decrease in rates contracts lending in the long run due to the compressed deposit spread. Recent literature has developed beta measures to quantify this sensitivity. Drechsler et al. (2017) first construct the deposit spread beta to measure banks' exposure to monetary policy and finds that this deposit spread beta is mainly driven by banks' market power in the deposit market. Drechsler et al. (2021) further construct banks' income beta and expense beta, showing that similar values between income beta and expense beta mean that banks closely match the interest rate sensitivities of their interest income and ex-

penses. These results suggest that maturity transformation helps banks hedge interest rate risk. However, both Drechsler et al. (2017) and Drechsler et al. (2021) construct constant betas, suggesting the sensitivity does not change over time. Unlike Drechsler et al. (2017) and Drechsler et al. (2021), Emin et al. (2023a) show that interest pass-through rates on deposits vary significantly with interest rates, which reduces the effectiveness of deposits as a hedge when interest rates increase. Consistent with Emin et al. (2023a), I also find that deposit beta changes over time. Distinct from existing literature, I focus on how the ICM within BHCs affect their subsidiary banks' deposit rate responses to monetary policy. I demonstrate that this response varies systematically with the federal funds rate—that is, the deposit beta is time-varying rather than constant.

Finally, this paper contributes to the broader literature on capital regulation by identifying the impact of parent debt on subsidiary banks' capital and the potential consequences. A vast majority of literature on bank capital regulation aims at mitigating banks' risk-taking behavior (e.g., Keeley (1990), Shrieves and Dahl (1992), Demsetz and Strahan (1997), and Ellul and Yerramilli (2013)). However, literature about parent debt and subsidiary banks' capital is scarce. Ashcraft (2008) finds that distressed affiliated banks within a multi-bank holding company are also more likely to receive capital injections from their parent company, recover more quickly, and are less likely to fail over the next year, suggesting that BHCs are a source of strength to their banking subsidiaries. My findings complement these insights by showing that parent debt can, to some extent, relax the capital constraints faced by subsidiary banks, as it indirectly supplements their capital through equity injections from the parent company. Moreover, this mechanism may undermine the effectiveness of capital regulation, since the capital supporting subsidiary banks is financed by parent debt rather than by subsidiary banks' new equity issuance.

3 Hypothesis development

In this section, I develop testable hypotheses regarding the economic mechanisms through which parent debt influences deposit beta. I focus on two primary mechanisms:

first, a *funding substitution effect*—I examine whether the parent debt is injected into the subsidiary bank as an alternative funding source; second, a *risk-mitigation effect*—I study whether parent debt can alter subsidiary banks' risk-taking, which in turn affects depositors' behavior and deposit pricing. The hypotheses below formalize these mechanisms:

3.1 Parent debt, internal capital transfer, and deposit beta

A central function of the ICM within BHCs is the strategic allocation of scarce capital among subsidiaries (Houston et al. (1997)). The subsidiary bank typically constitutes the bulk of total assets of the BHC. If parent-level financing flows directly into the bank, then a higher parent debt ratio may translate into increased internal capital transfer. This internally supplied funding serves as an alternative to traditional deposits. By relying on parent-sourced capital, subsidiary banks reduce their dependence on retail deposit markets to fund asset growth or maintain liquidity. This weakened reliance grants subsidiary banks greater flexibility in their pricing strategy; specifically, subsidiary banks face less pressure to aggressively raise deposit rates to retain funding when market rates climb. By easing the banks' external funding constraints, these internal transfers should reduce deposit beta.

Hypothesis 1. A higher parent debt ratio is associated with a lower deposit beta for the subsidiary bank, driven by increased internal capital transfers that facilitate a funding substitution effect.

3.2 Parent debt, risk-taking, and deposit beta

The second mechanism rests on the disciplinary role of parent debt. Ma et al. (2026) suggests that parent debt can reduce subsidiary risk-taking due to the interplay of limited liability and cross-guarantee provisions. When a subsidiary bank's risk decreases, depositors—who typically monitor bank health—require lower interest rate premiums (Kishan and Opiela (2012); Martinez Peria and Schmukler (2001)). If increased parent debt effectively curbs the subsidiary's risk-shifting incentives, subsidiary banks pay a smaller risk-weighted premium on their liabilities. As a result, deposit rates become less sensitive to fluctuations in the federal funds rate. In sum, by limiting subsidiary

banks' risk, parent debt reduces the intensity of depositor discipline and the necessity of high-beta rate adjustments. This safer profile stabilizes the relationship between market rates and deposit pricing.

Hypothesis 2. A higher parent debt ratio is associated with a lower deposit beta for the subsidiary bank when such debt reduces subsidiary banks' risk-taking, supporting a risk-mitigation effect.

4 Data, key Variables, and the estimation of deposit beta

In this section, I describe the data sample, the construction of the banks' parent debt ratio and deposit beta, and provide summary statistics. Table 1 presents detailed definitions of these variables.

4.1 Data sources

4.1.1 Subsidiary bank-level data

The subsidiary bank-level data are taken from the Consolidated Report of Condition and Income database (Call Reports), covering the period from January 1984 to December 2020. This dataset contains quarterly observations of income statements and balance sheets of all U.S. commercial banks, some of which are affiliated with BHCs. The dataset includes a unique bank-level identifier (RSSD9348), which allows for linkage to the corresponding BHC-level data.

4.1.2 BHC-level data

The BHC-level data are obtained from the Consolidated Financial Statements for Holding Companies (FR-Y9C) and the Parent Company Only Financial Statements for Large Holding Companies (FR-Y9LP). I use quarterly data spanning 1984 to 2020, which include balance sheets and income statements for all U.S. BHCs and their parent-only entities.

4.1.3 Fed funds data

I collect monthly effective federal funds rate data from the Federal Reserve Economic Data (FRED) for the period 1983–2020. Following Drechsler et al. (2023), I use the rate at the last month of each quarter to proxy for the quarterly effective federal funds rate.

4.2 Parent debt ratio

The key independent variable is the parent debt ratio ($ParentDebtRatio$), calculated as the parent's non-deposit liabilities divided by the BHC's total non-deposit liabilities. The parent's non-deposit liability includes the parent company's securities sold under agreements to repurchase, borrowings of commercial paper and other borrowings with a remaining maturity of one year or less, other borrowed money with a remaining maturity of more than one year, and subordinated notes and debentures. The BHC's non-deposit liability contains group-level consolidated total liabilities but excludes deposits in domestic offices, foreign offices, Edge and Agreement subsidiaries, and international banking facilities (IBFs). The parent debt ratio is winsorized at the 1% level. The time-series distribution of the parent debt ratio is shown in Figure 1. The mean quarterly parent debt ratio rises from below 4% to more than 12% after the global financial crisis.

4.3 Estimate deposit beta

The key dependent variable is the deposit beta, which measures the sensitivity of a bank's interest expense on deposits to the federal funds rate. In this section, I estimate deposit beta. Following Drechsler et al. (2021), I restrict the sample to banks with at least 60 quarterly observations between 1984 and 2020. I begin the analysis by running the following time-series regression for each subsidiary bank i in each quarter t:

$$\Delta y_{it} = \alpha_i + \eta_t + \sum_{\tau=0}^{3} \beta_{i,\tau} \Delta FedFunds_{t-\tau} + \epsilon_{it} \tag{1}$$

where Δy_{it} is the change in the deposit rate from $t-1$ to t for subsidiary bank i, $\Delta FedFunds_{t-\tau}$ is the change in the Fed funds rate from $t-\tau-1$ to $t-\tau$, α_i

are subsidiary bank fixed effects, and η_t are quarter fixed effects. The deposit rate is defined as total quarterly interest expense on deposits divided by the quarterly average asset value, and then annualized (multiplied by four). Following Emin et al. (2023b), I winsorize the deposit rate at the 1% level. I allow for three lags of the federal funds rate changes to capture the cumulative effect over a full year. To avoid look-forward bias, I apply a rolling-window estimation using the preceding 60 quarters of data to compute the deposit beta for the current quarter. The estimate of subsidiary bank i's overall deposit beta is the sum of the coefficients in (1), that is, $\beta_i = \sum_{\tau=0}^{3} \beta_{i,\tau}$, denoted by $DepositBetaTime$. Following Drechsler et al. (2021) and Emin et al. (2023b), I winsorize the deposit beta at the 5% level to minimize the impact of outliers. This procedure limits the main analysis period to 1999–2020.

Figure 2 shows the cross-sectional distribution of deposit betas, with a mean of 0.356, indicating that deposit rates adjust less than one-for-one to changes in the federal funds rate across commercial banks. Figure 3 plots the quarterly average deposit beta over time, showing a gradual decline.

4.4 Control variables in the baseline regression

I include several subsidiary bank-specific controls that are commonly found to influence deposit beta. These variables include subsidiary banks' size ($LnAT$), return on assets (ROA), and the equity ratio ($EquityRatio$). Kishan and Opiela (2000) argues that bank asset size and capital affect a bank's ability to raise funds and maintain loan growth during contractionary monetary policies.

4.5 Funding-substitution mechanism variables

To measure the internal capital transfer from the parent company to subsidiary banks, I construct two measures. One is $EquityInvestment$, and the other is $TotalInvestment$. $EquityInvestment$ is the equity investment from the parent company to subsidiary banks as a share of the BHC's total assets, while $TotalInvestment$ is the equity and debt investment from the parent company to subsidiary banks as a share of the BHC's total assets. $DepositRatio$ is a control variable to measure the deposit size within a BHC.

4.6 Risk-mitigation mechanism variables

To measure subsidiary banks' risk-taking, I use $Zscore$ to measure subsidiary banks' risk, defined as the return on assets plus the equity-to-asset ratio, scaled by the standard deviation of return on assets, where the equity-to-asset ratio is calculated as (book equity)/(total assets). Thus, $Zscore$ comprises a risk measure from both the liability and asset sides. A higher $Zscore$ indicates greater bank stability. Due to the skewness of $Zscore$, I use its natural logarithm, which follows a normal distribution (Laeven and Levine (2007)). Following Li and Song (2022), I also use the risk-weighted leverage ($RiskLeverage$), defined as a bank's risk-weighted assets divided by tier-1 capital, as the risk-taking measure. The risk-weighted leverage is the product of a bank's asset risk weight ($RiskWeight$) and leverage ($Leverage$), and thus summarizes a bank's risk-taking on both margins. Therefore, I also include both $RiskWeight$ and $Leverage$ as risk-taking measures. Furthermore, I include the Leverage Ratio ($LeverageRatio$) to measure subsidiary banks' risk-taking.

4.7 Summary statistics

Table 2, Panel A, reports the summary statistics of BHC-level variables. On average, the parent debt ratio is 5.53%, and more than half of the parent companies of BHCs do not issue any external debt. The mean value of $EquityInvestment$ is 8.86%. The mean value of $TotalInvestment$ is 8.92%. The BHC's mean deposit ratio is 80.3%.

Table 2, Panel B, reports the summary statistics of subsidiary bank-level variables. The mean value of deposit beta ($DepositBetaTime$) is 0.36. The average logarithm of asset size ($LnAT$) is 13.07, which translates to a mean of USD 0.47 billion in total assets. Subsidiary banks exhibit an average ROA (ROA) of 0.69% and an equity ratio ($EquityRatio$) of 10.4%. For the risk-taking measures, the averages of Leverage, LeverageRatio, RiskLeverage, RiskWeight, and Zscore are 11.68, 0.96, 8.24, 0.71, and 3.41, respectively.

Table 2, Panel C, reports the summary statistics of macro-level variables. The averages of MonetaryPolicySurprise (MPS) and the quarterly change in the federal funds rate ($FedfundsChange$) are -0.24% and 0.04%, respectively.

5 Empirical analysis and results

This section presents empirical results on the impact of the parent debt ratio on deposit beta, and on the mechanisms through which parent debt impacts deposit beta. Specifically, I examine how a funding substitution effect and a risk-mitigation effect affect the impact of the parent debt ratio on deposit beta.

5.1 The impact of parent debt on deposit beta

In this section, I conduct the main analysis by examining the effect of the parent debt ratio on its subsidiary banks' deposit beta. I merge subsidiary bank-level data with the corresponding BHC-level data using bank identifiers. The final sample consists of 2,479 BHCs and 4,168 subsidiary banks.

For each BHC-subsidiary bank pair, I estimate the following model:

$$\begin{aligned} DepositBetaTime_{i,j,t} = \alpha + \beta ParentDebtRatio_{j,t-1} \\ + \gamma Controls_{i,t-1} + \lambda_i + \eta_t + \epsilon_{ijt} \end{aligned} \tag{2}$$

where, for each subsidiary bank i affiliated with BHC j in quarter t, $DepositBetaTime$ is the deposit beta estimated in Section 4.3. $ParentDebtRatio$ is defined as the parent's non-deposit liabilities divided by the BHC's total non-deposit liabilities. $Controls$ denote a set of subsidiary bank-level control variables described in Section 4.5 and known to be relevant to deposit beta. λ_i is the subsidiary bank fixed effect that absorbs invariant bank characteristics. η_t is the quarter fixed effect to capture time-varying characteristics. Standard errors are clustered at both the subsidiary bank and quarter levels. When I interpret the regression results, I focus on the sign and significance level of β.

Table 3 presents evidence of a significant negative relation between the parent debt ratio and deposit beta. Model (1) presents the unconditional regression, where the coefficient on the parent debt ratio is negative and statistically significant at the 1% level, suggesting that subsidiary banks experience lower deposit betas when their parent companies issue more external debt. This effect remains robust after including subsidiary bank-specific controls in Model (2). Specifically, a one standard deviation increase in

the parent debt ratio (0.112) corresponds to an expected 0.23% decrease in the deposit beta (-0.0208 x 0.112 x 100% = -0.23%). These results indicate that higher parent debt ratios reduce subsidiary banks' deposit beta, supporting the view that the ICM within BHCs help subsidiary banks mitigate the impact of monetary policy on deposit beta.

With respect to the control variables, the subsidiary bank's asset size is positively associated with the deposit beta. Subsidiary banks with larger asset holdings tend to exhibit greater sensitivity to changes in the federal funds rate. In addition, the subsidiary bank's return on assets is positively related to the deposit beta. However, I find no significant evidence that the subsidiary bank's equity ratio affects the deposit beta.

5.2 Identification

Table 3 shows a negative relationship between the parent debt ratio and subsidiary banks' deposit betas, but this correlation may be spurious. A first-order concern is that the parent debt ratio may not be exogenous due to the omitted variables problem: that is, a BHC may endogenously determine its parent debt ratio based on characteristics unobservable to the econometrician. For example, a BHC under regulatory pressure to improve capital ratios may issue more parent debt to inject capital into subsidiaries. If such activities also correlate with deposit pricing strategies, OLS estimates would be biased. Therefore, estimation of the impact of the parent debt ratio on deposit beta could be seriously biased if endogeneity in the parent debt ratio is not appropriately addressed.

While it is very difficult to address the omitted variable concern in the absence of a natural experiment, I control for endogeneity in the parent debt ratio by constructing an instrumental variable and conducting 2SLS regressions within the limitations of the data.

5.2.1 The instrumental variable approach

The instrument for the parent debt ratio is the monetary policy surprise (*MPS*) constructed by Bauer and Swanson (2023). This variable measures intraday interest rate changes over a 30-minute window starting 10 minutes before each FOMC announce-

ment and ending 20 minutes afterward, using high-frequency data from Tick Data. According to Bauer and Swanson (2023), a hawkish surprise is denoted by $MPS > 0$ while a dovish surprise is denoted by $MPS < 0$. In the empirical analysis, the surprise is standardized so that a one-unit increase corresponds to a 25-basis-point movement in the 1-year Treasury yield during the sample period.

The rationale behind this instrumental variable is threefold. First, changes in interest rates directly influence the cost of BHC parent debt. A hawkish surprise makes issuing new parent debt more expensive while a dovish surprise make it cheaper. By affecting debt issuance decisions following FOMC announcements, the proposed instrument satisfies the relevance requirement. Second, because the monetary policy surprise is unanticipated by the market, it is exogenous to the BHC's immediate financial decisions; specifically, it is impossible for parent debt issuance decisions to influence the monetary policy surprise itself (see also Kuttner, 2001; Gürkaynak et al., 2005). Third, the instrument, to some extent, satisfies the exclusion restriction. The deposit beta in the current quarter is estimated using a rolling window of the previous sixty quarters, and the $ParentDebtRatio$ is one-quarter lagged in the first- and second-stage regressions. Therefore, the evidence suggests that monetary policy surprises do not directly affect the deposit beta; rather, their impact operates indirectly through changes in the parent debt ratio. If the intuition behind the instrument construction is supported, I expect to observe a significantly negative coefficient estimate of the instrument in the first-stage regressions.

Table 4, Model (1), presents the results of the first-stage regression with the parent debt ratio as the dependent variable. Consistent with the intuition for the instrument, the coefficient estimate for MPS is negative and statistically significant at the 10% level, suggesting that a positive monetary policy surprise reduces the parent debt ratio. Since the t-statistic of the instrument is significant, the instrument is highly correlated with the parent debt ratio. Based on the rule of thumb for one instrument (for one endogenous variable), I reject the null hypothesis that the instrument is weak. Consequently, the second-stage coefficient estimates are likely to be consistent, and the associated standard errors provide a reliable basis for statistical inference.

A primary concern in this identification strategy is the potential overlap between

monetary policy surprises and broader macroeconomic trends. To isolate the exogenous component of the monetary policy surprise and mitigate omitted-variable bias, I include one- to four-quarter lags of key macroeconomic variables: the one-year Treasury rate (T1yr), the logarithm of industrial production (LnIP), the logarithm of the producer price index (LnPPI), the logarithm of commodity prices (LnCP), the excess bond premium (EBP), and the logarithm of real GDP (LnRGDP) (Gilchrist and Zakrajšek (2012)). By controlling for these lagged variables, the model accounts for the prevailing economic environment that might otherwise confound the relationship between the parent debt ratio and deposit pricing.

Before discussing the regression results, Table 2, Panel C, presents the summary statistics of the macro-level variables used in the 2SLS analysis. Over the 1999–2020 period, the average values of the macro-level control variables—T1yr, LnIP, LnPPI, LnCP, EBP, and LnRGDP—are 2.51, 94.26, 160.8, 162.9, 0.142, and 9.671, respectively.

In the second stage, I regress the deposit beta on the fitted values of the parent debt ratio obtained from the first-stage estimation. To ensure the robustness of the results, the model includes a comprehensive suite of subsidiary bank-level controls and subsidiary bank fixed effects. Table 4, Model (2), reports the second-stage estimates. The coefficient for the parent debt ratio is negative and statistically significant at the 1% level, providing empirical support for the hypothesis that a higher parent debt ratio significantly dampens deposit beta. According to the estimate in Model (2) of Table 4, a one standard deviation increase in the parent debt ratio (0.112) corresponds to an expected 0.17% decrease in the deposit beta (-0.0156 x 0.112 x 100% = -0.17%).

Comparing the OLS results (Table 3, Model (2)) with the 2SLS estimates (Table 4, Model (2)), it is noteworthy that the magnitude of the 2SLS coefficient estimate is smaller than that of the OLS estimate, even though the coefficient estimates from both approaches are negative and statistically significant. This pattern suggests that the OLS regression biases the coefficient estimate downward due to endogeneity in the parent debt ratio. This finding suggests that omitted variables create a spurious additional negative correlation that makes OLS overestimate the magnitude of the negative effect. For instance, a BHC under regulatory pressure to improve capital ratios for its subsidiary banks issues more parent debt to inject capital into subsidiary banks, while

the subsidiary bank facing the same regulatory pressure becomes more conservative with a lower deposit beta to reduce risk and shrink balance sheets. This observation does not necessarily reflect a causal consequence of the parent debt ratio on subsidiary banks' deposit beta because of the endogeneity problem inherent in the optimal choice for parent debt. However, the negative correlation caused by the omitted variable is the main driving force that biases the coefficient estimate of the parent debt ratio downward. Once I use the instrument to clean up the correlation between the BHC's parent debt choice and the residuals (the subsidiary bank's unobservable characteristics) in the structural equations, the endogeneity of the BHC's optimal parent debt choice is removed and the coefficient estimates increase, that is, become less negative.

In summary, the identification tests based on the instrumental-variable approach reported in this section indicate a negative causal effect of the parent debt ratio on deposit beta.

5.3 Parent debt, internal capital transfer, and deposit beta

In this subsection, I examine how parent debt affects subsidiary banks' deposit beta through the first proposed economic mechanism: a funding substitution effect. I examine whether internal transfers from parent companies to subsidiary banks exist. Specifically, I analyze the relationship between the parent debt ratio and the parent's equity investment or total investment (equity and debt investment) in its subsidiary banks. However, some BHCs have multiple subsidiary banks, but the available data do not report investment amounts from the parent company to each subsidiary bank within the same BHC, and I can only observe the total investment amount from the parent company to all its subsidiary banks. Therefore, I conduct the analysis at the BHC level, using aggregate investment measures. I estimate the following model:

$$\begin{aligned} Investment_{jt} = \alpha + \beta ParentDebtRatio_{j,t-1} + \\ \gamma Controls_{j,t-1} + \lambda_j + \eta_t + \epsilon_{jt} \end{aligned} \tag{3}$$

where $Investment_{jt}$ is either $EquityInvestment$ or $TotalInvestment$. $Controls$ represent BHC-level control variables described in Section 4.6 and are lagged by one

quarter to mitigate concerns over look-ahead bias. All models include BHC and quarter fixed effects, and standard errors are clustered at both the BHC and quarter levels.

Table 5 presents the main results of my BHC-level analysis. The analysis begins by using $EquityInvestment$ as the dependent variable. Model (1) demonstrates that for a one-unit increase in the parent debt ratio, equity investment increases 0.62 percentage points. The results on equity investment are robust after accounting for the deposit ratio of the BHC (Model (2)). In addition to equity investment, I also consider whether a high parent debt ratio increases $TotalInvestment$ in its bank subsidiaries. Models (3) and (4) of Table 5 confirm that high parent debt ratios are also associated with high total investment in subsidiary banks.

Next, I conduct the analysis by examining the effect of internal capital transfer on subsidiary banks' deposit beta. For each BHC-subsidiary bank pair, I estimate the following model:

$$DepositBetaTime_{i,j,t} = \alpha + \beta Investment_{j,t-1} \\ \gamma Controls_{i,t-1} + \eta \Delta FedFunds_{t-1} + \lambda_i + \epsilon_{ijt} \qquad (4)$$

where, for each subsidiary bank i affiliated with BHC j in quarter t, $Investment$ is either $EquityInvestment$ or $TotalInvestment$. $Controls$ denote a set of subsidiary bank-level control variables described in Section 4.5 and known to be relevant to deposit beta. Standard errors are clustered at both the subsidiary bank and quarter levels. I include subsidiary bank fixed effects to mitigate omitted-variable bias and the change in the federal funds rate to capture the business cycle that changes with time.

Table 6 presents the results of my empirical analysis. Models (1) and (2) use $EquityInvestment$ as the key independent variable. Model (2) shows that the coefficient on $EquityInvestment$ is negative and statistically significant at the 5% level, suggesting that subsidiary banks experience lower deposit betas when their parent companies inject more capital in the form of equity into subsidiary banks. Specifically, a one standard deviation increase in $EquityInvestment$ (0.034) corresponds to an expected 0.46% decrease in the deposit beta (-0.134 $\times$ 0.034 $\times$ 100% = -0.46%). Models (3) and (4) use $TotalInvestment$ as the key independent variable. Model (4) shows that the

coefficient on $TotalInvestment$ is negative and statistically significant at the 5% level, and a one standard deviation increase in $TotalInvestment$ (0.034) corresponds to an expected 0.44% decrease in the deposit beta (-0.128 × 0.034 × 100% = -0.44%).

The results indicate that higher internal capital transfers reduce subsidiary banks' deposit beta, supporting the view that the ICM within BHCs help subsidiary banks mitigate the impact of monetary policy transmission by transferring capital from the parent company to subsidiary banks.

Overall, the above results confirm that parent companies channel parent debt into their subsidiary banks mainly through equity investments via the ICM. Together, the BHC- and subsidiary-level evidence suggests that parent companies primarily inject debt-funded capital into subsidiary banks mainly in the form of equity investment, providing an alternative funding source for subsidiary banks, which reduces subsidiary banks' reliance on deposits and decreases deposit beta. The empirical evidence thus supports *Hypothesis 1*.

5.4 Parent debt and subsidiary banks' risk-taking

In this subsection, I examine how parent debt affects subsidiary banks' deposit beta through the second proposed economic mechanism: a risk-mitigation effect. For each subsidiary bank, I estimate the following model:

$$\begin{aligned} RiskTaking_{ijt} = \alpha + \beta ParentDebtRatio_{j,t-1} + \\ \gamma Controls_{i,t-1} + \eta \Delta FedFunds_{t-1} + \lambda_i + \epsilon_{ijt} \end{aligned} \tag{5}$$

where, for each subsidiary bank i affiliated with BHC j in quarter t, $RiskTaking$ is one of the risk-taking measures of subsidiary banks. $Controls$ represents a set of bank-level control variables described in Section 4.5. All explanatory variables are lagged by one quarter to mitigate concerns over look-ahead bias. In all models, standard errors are clustered at both subsidiary bank and quarter levels. I control for the quarterly change in the federal funds rate to capture the business cycle and include subsidiary bank fixed effects to absorb bank characteristics that do not change with time.

Table 7 presents the empirical evidence showing a negative relationship between

the parent debt ratio and subsidiary banks' risk-taking during the 1999-2020 period. Specifically, for a one standard deviation increase in the parent debt ratio (0.118), I observe an expected 1.14% increase in Z-Score (0.0964 x 0.118 x 100% = 1.14%), a 3.36% decrease in risk-weight leverage (-0.2847 x 0.118 x 100% = -3.36%), and a 3.40% decrease in leverage ratio (-1.2881 x 0.118 x 100% = -3.40%).

Next, I conduct the analysis by examining the effect of subsidiary bank risk-taking on deposit beta. For for each subsidiary bank i in quarter t , I estimate the following model:

$$DepositBetaTime_{i,t} = \alpha + \beta RiskTaking_{i,t-1} + \gamma Controls_{i,t-1} + \eta \Delta FedFunds_{t-1} + \lambda_i + \epsilon_{it} \quad (6)$$

where $RiskTaking$ denotes one of the subsidiary bank risk-taking measures described previously. $Controls$ represents a vector of subsidiary bank-level characteristics outlined in Section 4.5, selected on the basis of their established relevance to deposit beta. All specifications include subsidiary bank fixed effects to address omitted-variable bias, and the change in the federal funds rate to account for time-varying macroeconomic conditions. Standard errors are two-way clustered at the subsidiary bank and quarter levels.

Table 8 reports the estimation results. Models (1) and (2) employ $Zscore$ as the key independent variable. Model (2) shows that the coefficient on $Zscore$ is negative and statistically significant at the 1% level, indicating that subsidiary banks with lower overall risk exhibit lower deposit betas. Specifically, a one standard deviation increase in $Zscore$ (0.575) is associated with a 0.64% decrease in the deposit beta (-0.0112 $\times$ 0.575 $\times$ 100% = -0.64%).

Models (3) and (4) use $RiskLeverage$ as the key independent variable. Model (4) shows that the coefficient on $RiskLeverage$ is positive and statistically significant at the 1% level. A one standard deviation increase in $RiskLeverage$ (2.123) corresponds to an expected 1.4% increase in the deposit beta (0.0066 $\times$ 2.123 $\times$ 100% = 1.4%).

Models (5) and (6) use $RiskWeight$ as the key independent variable. The coefficient in Model (6) is positive and statistically significant at the 1% level, though the eco-

nomic magnitude is modest: a one standard deviation increase in $RiskWeight$ (0.127) corresponds to an expected 0.08% increase in the deposit beta ($0.0066 \times 0.127 \times 100\% = 0.08\%$).

Models (9) and (10) employ *Leverage* as the primary independent variable. Model (10) shows that the coefficient on *Leverage* is positive and statistically significant at the 1% level, and a one standard deviation increase in *Leverage* (2.466) corresponds to an expected 16.42% increase in the deposit beta ($0.0036 \times 2.466 \times 100\% = 0.89\%$).

Taken together, these results consistently indicate that lower subsidiary bank risk is associated with reduced deposit betas, supporting the view that the ICM within the BHC enable subsidiary banks to attenuate the pass-through of monetary policy to deposit rates by curbing excessive risk-taking.

Overall, the results presented in this subsection support *Hypothesis 2*. The evidence suggests that parent debt constrains subsidiary bank risk-taking, which in turn dampens the sensitivity of subsidiary deposit rates to changes in the monetary policy rate.

5.5 Robustness tests

In this section, I examine the robustness of my main findings. I first test whether the documented effects of the parent debt ratio on deposit beta remain robust under an alternative method for calculating deposit beta. Next, I assess whether my results hold when using alternative proxies for deposit beta. I report the estimation results of Equation (2). Overall, I find qualitatively consistent results when using alternative methods for estimating deposit beta or using alternative proxies for deposit beta—expense beta as the dependent variable.

5.5.1 Alternative calculations of deposit beta

Since my dependent variable, $DepositBetaTime$, is an estimated measure, I test whether my baseline results are driven by the estimation method. To do so, I re-estimate the sensitivity of a subsidiary bank's deposit rate to the federal funds rate using an alternative specification that excludes time fixed effects, denoted by $DepositBeta$. Specifically, I estimate the following model:

$$\Delta y_{it} = \alpha_i + \sum_{\tau=0}^{3} \beta_{i,\tau} \Delta FedFunds_{t-\tau} + \epsilon_{it} \quad (6)$$

This specification is identical to Equation (1), except that it omits quarter fixed effects. Table 1, Panel A, shows that the mean value of *DepositBeta* estimated from this specification is 0.367, which is slightly higher than the deposit beta estimated by including quarter fixed effects.

I then re-estimate Equation (2) using *DepositBeta* as the dependent variable. The coefficient estimates of the parent debt ratio, reported in Table 9, Model (2), are negative and significant at the 1% level. According to the estimate reported in Model (2) of Table 9, subsidiary banks' deposit beta decreases by, on average, 0.02 units for a one-unit increase in the parent debt ratio. The evidence is consistent with the main results in Table 3, and confirms that a higher parent debt ratio is associated with a lower deposit beta. This finding indicates that my baseline results are not driven by the specific estimation method used to calculate deposit beta.

To address the endogeneity issue, I also instrument the parent debt ratio with MPS and re-run Equation (2). Table 10, Model (2), reports the second-stage result, which shows that when the parent debt ratio increases one unit, deposit beta decreases 0.012 units.

Overall, the above results confirm that the main result is not driven by the specific estimation method used to calculate deposit beta.

5.5.2 Alternative proxies for deposit beta

To further test robustness, I follow Drechsler et al. (2021) and construct $ExpenseBetaTime$ as an alternative proxy for interest rate sensitivity. $ExpenseBetaTime$ measures the sensitivity of a bank's interest expense rate to changes in the federal funds rate. Specifically, I construct the interest expense rate as the total interest expense divided by the quarterly average assets (multiplied by four), and then use Equation (1) to regress changes in this rate on changes in the federal funds rate to obtain $ExpenseBetaTime$.

I use this $ExpenseBetaTime$ as the dependent variable and re-run Equation (2).

The results, reported in Models (1) and (2) of Table 11, continue to show a negative and statistically significant relationship between the parent debt ratio and expense beta. When I estimate the expense beta using Equation (6) by replacing the change in deposit rate with the change in expense rate, and re-run the regression using Equation (2) with the expense beta estimated without time fixed effects as the dependent variable, denoted by $ExpenseBeta$, the results, reported in Models (3) and (4) of Table 11, show that the coefficient on the parent debt ratio remains negative and significant at the 10% level, further confirming the robustness of my findings.

One potential concern is endogeneity in the relationship between the parent debt ratio and expense beta. To address this, I replicate the 2SLS analysis in Section 5.2, using MPS constructed by Bauer and Swanson (2023) as an instrument for the parent debt ratio. The second-stage regression results are presented in Table 12.

Model (2) of Table 12 reports the second-stage estimates of Equation (2), replacing the parent debt ratio with its fitted value from the first stage and using $ExpenseBetaTime$ as the dependent variable. The coefficient on parent debt remains negative and significant at the 1% level. Model (3) of Table 12 reports results using $ExpenseBeta$ as the dependent variable, which also yields a negative and statistically significant coefficient at the 1% level, reinforcing my baseline findings.

In summary, the instrumental variable analysis presented in this section provides consistent evidence of a negative causal effect of the parent debt ratio on expense beta. Together, these robustness checks confirm that my main findings are not driven by the estimation method or the specific proxy used to measure banks' interest rate sensitivity.

6 A simple model

I use a simple model to illustrate the regulatory concern: how parent debt allows subsidiary banks to circumvent capital requirements. Suppose a bank holding company (BHC) consists of a parent company, a bank subsidiary, and a non-bank subsidiary. The BHC (on a consolidated basis) and the bank subsidiary must satisfy a 10% capital requirement, while the parent company and the non-bank subsidiary do not face capital requirements. The parent company wholly owns both the bank subsidiary and the non-

bank subsidiary.

6.1 $t = 0$

At $t = 0$, the consolidated balance sheet of the BHC comprises 101 units of assets, 90 units of debt, and 11 units of equity. The parent comp
any invests 10 units of equity in the bank subsidiary and 1 unit of equity in the non-bank subsidiary. The balance sheets at $t = 0$ are as follows:

BHC's Balance Sheets at $t = 0$

BHC

101	90(D)
	11(E)

Parent

10	0(D)
1	11(E)

Bank Subsidiary

100	90(D)
	10(E)

Non-bank Subsidiary

1	0(D)
	1(E)

6.2 $t = 1$

At $t = 1$, the parent company issues 1 unit of parent debt and invests it as equity in its bank subsidiary. The bank subsidiary uses this additional capital to take more deposits. Since the bank subsidiary must maintain a 10% capital ratio, the maximum additional deposits it can take is 9 units. However, the BHC must also satisfy the 10% capital ratio requirement on a consolidated basis. Taking 9 units of additional deposits would cause the BHC to breach its capital requirement. To satisfy the capital requirement at both the subsidiary bank and BHC levels, the maximum deposits the subsidiary bank can take is 8 units. The balance sheets at $t = 1$ are as follows:

BHC's Balance Sheets at $t = 1$

BHC

110	99(D)
	11(E)

Parent Company

11	1(D)
1	11(E)

Bank Subsidiary

109	98(D)
	11(E)

Non-bank Subsidiary

1	0(D)
	1(E)

This model demonstrates that when the parent company issues external debt and invests it in its subsidiary bank as equity, it effectively supplements the subsidiary bank's capital. This means that part of the subsidiary bank's capital originates from parent debt rather than from the subsidiary bank's own equity issuance. This capital

injection enables the subsidiary bank to take more deposits and enlarges the balance sheet sizes of both the subsidiary bank and the BHC.

7 Conclusion

This paper provides evidence on how parent debt affects subsidiary banks' deposit betas through the ICM. Using a comprehensive dataset covering all domestic U.S. commercial banks and their corresponding BHCs from 1984 to 2020, combined with an instrumental variable identification strategy, I identify two economic mechanisms linking parent debt to subsidiary banks' deposit betas. My findings show that, for subsidiary banks affiliated with BHCs, the ICM facilitate capital transfers from parent companies to their subsidiary banks. The capital provided by the parent is primarily sourced from the parent's external debt. This internal capital transfer reduces subsidiary banks' reliance on deposit funding, thereby granting them greater flexibility in deposit-rate setting and lowering their sensitivity to changes in the federal funds rate. Moreover, higher parent debt ratio—often subject to cross-guarantee provisions—incentivizes the parent to curb subsidiary bank's risk-taking. As subsidiary banks' risk decreases, depositors demand a lower risk premium, further decoupling deposit rates from policy rate fluctuations.

A series of robustness checks confirms that the results are consistent across alternative measures and proxies of deposit beta. Overall, the study offers new insights into the real effects of the ICM within BHCs on the deposit beta of their subsidiary banks.

Furthermore, my findings raise an important regulatory concern regarding subsidiary banks' capital adequacy. When parent companies issue external debt and inject the proceeds into subsidiary banks in the form of equity, they effectively provide an alternative source of capital that allows subsidiary banks to supplement their regulatory capital with parent debt. If the amount of parent debt channeled to subsidiary banks exceeds the required capital thresholds, this double leverage may enable subsidiary banks to circumvent capital regulations, potentially undermining the stability of the banking sector.

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Table 1: Variable definitions. This table summarizes the variable definitions in our study.

Variable	Definition
DepositBeta	A measure of the sensitivity of changes in deposit interest expense to changes in the federal funds rate. Estimated without time fixed effects.
DepositBetaTime	A measure of the sensitivity of changes in deposit interest expense to changes in the federal funds rate. Estimated including both time and individual fixed effects.
ExpenseBeta	A measure of the sensitivity of changes in interest expense to changes in the federal funds rate. Estimated without time fixed effects.
ExpenseBetaTime	A measure of the sensitivity of changes in interest expense to changes in the federal funds rate. Estimated including both time and individual fixed effects.
EquityInvetsment	EquityInvetsment is the ratio of parent equity investment in its subsidiary banks to the BHC's total assets.
TotalInvestment	TotalInvestment is the ratio of parent equity and debt investment in its subsidiary banks to the BHC's total assets.
EquityRatio	The ratio of equity to total assets.
EBP	Excess bond premium.
FedfundsChange	FedfundsChange is the quarterly change of the federal funds rate.
Leverage	Leverage is measured by (Total assets)/(Tier 1 capital).
LeverageRatio	LeverageRatio is tier 1 leverage capital ratio.
LnCP	The logarithm of commodity prices.
LnRGDP	The logarithm of real GDP.
LnIP	The logarithm of business-sector industrial production.
LnPPI	The logarithm of the Finished Goods Producer Price Index (PPI).
Ln AT	The logarithm of the subsidiary bank's total assets.
MonetaryPolicySurprise (MPS)	High-frequency changes in interest rates around monetary policy announcements, as constructed by Bauer and Swanson (2023). The surprise is standardized so that a one-unit increase corresponds to a 25-basis-point movement in the 1-year Treasury yield during the sample period.

ParentDebtRatio	Parent debt ratio is the ratio of the immediate parent's non-deposit debt liabilities to BHC's non-deposit liability. It is estimated as quarterly (BHCP2309(parent company's borrowings of commercial paper with a remaining maturity of one year or less) + BHCP2332(parent company's other borrowings with a remaining maturity of one year or less) + BHCP0368(parent company's other borrowed money with a remaining maturity of more than one year) + BHCP4062(parent company's subordinated notes and debentures)+ BHCP0279(parent company's securities sold under agreements to repurchase))/(BHCK2948(group-level consolidated total liabilities)-BHDM6631(group-level consolidated noninterest-bearing deposits in domestic offices)-BHDM6636(group-level consolidated interest-bearing deposits in domestic offices)-BHFN6631 (group-level consolidated noninterest-bearing deposits in foreign offices, Edge and Agreement subsidiaries, and IBFs)-BHFN6636 (group-level consolidated interest-bearing deposits in foreign offices, Edge and Agreement subsidiaries, and IBFs))
RiskLeverage	Risk leverage is the ratio of risk-weighted assets to tier 1 capital.
RiskWeight	Risk weight is measured by (Total risk-weighted assets)/(Total asset).
ROA	ROA is estimated as net income divided by total assets.
T1yr	1-Year Treasury rate.
Zscore	Z-score is a proxy for bank stability. It is estimated as return on assets plus equity-to-asset ratio, scaled by the standard deviation of return on assets, where equity-to-asset ratio is estimated by (book equity)/(total assets). The variable is calculated at the annual level. To avoid the skewness, I take the logarithm of it.

Table 2: Summary statistics.

Panel A reports the BHC-level averages, including the pooled averages of ParentDebtRatio, EquityInvestment, TotalInvestment, and DepositRatio. The sample period is 1999-2020. The sample consists of 2,479 unique BHCs, some of which have more than one bank subsidiaries.

VARIABLES	(1) N	(2) mean	(3) sd	(4) p10	(5) p25	(6) p50	(7) p75	(8) p90	(9) min	(10) max
ParentDebtRatio	84,806	0.0553	0.112	0	0	0	0.0586	0.191	0	0.584
EquityInvestment	84,684	0.0886	0.0342	0.0575	0.0775	0.0904	0.106	0.123	-0.0439	0.976
TotalInvestment	84,702	0.0892	0.0342	0.0585	0.0781	0.0910	0.106	0.124	-0.0439	0.976
DepositRatio	84,806	0.803	0.0889	0.704	0.768	0.821	0.861	0.887	0	0.998

Panel B reports the subsidiary bank-level averages, including the pooled averages of beta variables (DepositBeta, DepositBetaTime, ExpenseBeta, and ExpenseBetaTime), and the risk-taking measures (Leverage, LeverageRatio, RiskLeverage, RiskWeight, and Zscore). The sample period is 1999-2020, covering 4,168 unique subsidiary banks.

VARIABLES	(1) N	(2) mean	(3) sd	(4) p10	(5) p25	(6) p50	(7) p75	(8) p90	(9) min	(10) max
DepositBeta	126,960	0.367	0.112	0.232	0.299	0.366	0.435	0.507	0.0841	0.623
DepositBetaTime	126,960	0.356	0.112	0.223	0.288	0.355	0.423	0.497	0.0775	0.619
ExpenseBeta	126,960	0.387	0.108	0.262	0.322	0.381	0.445	0.523	0.116	0.655
ExpenseBetaTime	126,960	0.375	0.109	0.250	0.310	0.368	0.433	0.513	0.109	0.647
LnAT	126,960	13.07	1.571	11.35	12.09	12.86	13.82	14.98	7.137	21.83
EquityRatio	126,960	0.104	0.0614	0.0723	0.0812	0.0936	0.111	0.136	-0.0853	1
Leverage	126,958	11.68	2.466	8.634	10.15	11.70	13.19	14.64	4.994	18.68
LeverageRatio	106,949	0.963	2.626	0.0718	0.0797	0.0906	0.109	4.240	0.0566	11.17
RiskLeverage	126,835	8.240	2.123	5.419	6.896	8.391	9.745	10.72	2.411	13.80
RiskWeight	126,837	0.706	0.127	0.543	0.632	0.717	0.792	0.857	0.323	0.990
ROA	126,960	0.00694	0.0463	0.00158	0.00314	0.00605	0.00976	0.0135	-0.769	10.56
Zscore	125,591	3.405	0.575	2.797	3.072	3.355	3.686	4.106	1.657	5.271

Panel C reports the Macro-level averages, including the pooled averages. The sample period is 1999-2020.

VARIABLES	(1) N	(2) mean	(3) sd	(4) p10	(5) p25	(6) p50	(7) p75	(8) p90	(9) min	(10) max
EBP	126,960	0.142	0.735	-0.549	-0.373	-0.107	0.524	1.086	-0.771	3.358
FedfundsChange	126,960	-0.000380	0.00488	-0.00700	-0.000500	1.00e-04	0.00140	0.00550	-0.0165	0.00680
LnCP	126,960	162.9	14.62	148.5	150.3	157.4	176.0	185.9	142.5	188.8
LnIP	126,960	94.26	4.612	88.46	90.61	92.96	98.23	101.2	85.14	104.0
LnPPI	126,960	160.8	23.29	137.3	140.1	152.8	179.4	196.8	131.5	207.5
LnRGDP	126,960	9.671	0.107	9.556	9.573	9.671	9.736	9.826	9.497	9.952
MonetaryPolicySurprise	126,960	-0.00240	0.368	-0.472	-0.239	-0.00179	0.181	0.418	-1.891	1.201
T1yr	126,960	2.514	1.955	0.173	0.577	2.100	4.523	5.160	0.103	6.217

Table 3: Parent debt and deposit beta estimated with time fixed effects.

This table presents the results from our main analysis in Section 5.1. The dependent variable is the deposit beta estimated with time fixed effect, denoted by DepositBetaTime, and the key explanatory variable is the parent debt ratio. Model (1) only includes the parent debt ratio as the explanatory variable. Model (2) adds the full set of control variables. All models use quarterly data for 1999-2020. T-statistics reported in the paratheses are based on robust standard error clustered at the subsidiary bank and quarter levels. Superscripts ***, **, * indicates statistical significance at 1%, 5%, and 10% levels, respectively.

VARIABLES	(1) DepositBetaTime	(2) DepositBetaTime
ParentDebtRatio	-0.0207***	-0.0208***
	(0.0075)	(0.0074)
Ln AT		0.0130***
		(0.0034)
ROA		0.0368***
		(0.0056)
EquityRatio		-0.0019
		(0.0405)
Constant	0.3578***	0.1881***
	(0.0005)	(0.0447)
Observations	126,828	126,828
R-squared	0.7833	0.7846
Bank_Subsidiary FE	Yes	Yes
Quarter FE	Yes	Yes

Table 4: Relationship between parent debt and deposit beta estimated with time fixed effects - 2SLS analysis.

This table reports the results of the two-stage least squares (2SLS) estimation described in Section 5.2. The instrumental variable, MonetaryPolicySurprise (MPS), captures high-frequency changes in interest rates around FOMC announcements. Model (1) presents the first-stage results, and Model (2) reports the second-stage estimates. All models use quarterly data for 1999–2020. T-statistics reported in parentheses are based on robust standard errors clustered at the subsidiary bank level. Superscripts ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

VARIABLES	(1) Parent DebtRatio	(2) DepositBetaTime
LagMPS	-0.0016*	
	(0.0009)	
ParentDebtRatio		-0.0156***
		(0.0018)
Ln AT		0.0102***
		(0.0034)
ROA		0.0358***
		(0.0036)
EquityRatio		0.0302
		(0.0397)
Constant	0.0684***	-3.0631***
	(0.0004)	(0.2337)
Observations	121,997	108,367
R-squared	0.0000	0.8184
Macro-level Controls	No	Yes
Bank_Subsidiary FE	No	Yes
Quarter FE	No	No

Table 5: Parent debt and investment in subsidiary banks.

This table presents the results of the models described in Section 5.3. The dependent variable is equity investment or total investment (equity and debt investment), and the key explanatory variable is parent debt ratio. All models use quarterly data for 1999–2020 and include BHC and quarter fixed effects. T-statistics reported in parentheses are based on robust standard errors clustered at the BHC and quarter levels. Superscripts ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)
VARIABLES	Equity Investment	Equity Investment	Total Investment	Total Investment
ParentDebtRatio	0.0062**	0.0095***	0.0066**	0.0099***
	(0.0028)	(0.0029)	(0.0028)	(0.0029)
DepositRatio		-0.0470***		-0.0479***
		(0.0125)		(0.0125)
Constant	0.0884***	0.1259***	0.0890***	0.1273***
	(0.0002)	(0.0100)	(0.0002)	(0.0100)
Observations	81,618	81,618	81,636	81,636
R-squared	0.7591	0.7620	0.7574	0.7604
BHC	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes

Table 6: Investment in subsidiary banks and deposit beta estimated with time fixed effects

This table presents the results of the models described in Section 5.3. The dependent variable is DepositBetaTime, and the key explanatory variable is equity investment or total investment (equity and debt investment). All models use quarterly data for 1999-2020. T-statistics reported in the paratheses are based on robust standard error clustered at the subsidiary bank and quarter levels. Superscripts ***, **, * indicates statistical significance at 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)
VARIABLES	DepositBetaTime	DepositBetaTime	DepositBetaTime	DepositBetaTime
Equity Investment	-0.5305***	-0.1343**		
	(0.0687)	(0.0593)		
Total Investment			-0.5160***	-0.1283**
			(0.0686)	(0.0605)
Ln AT		-0.0343***		-0.0344***
		(0.0034)		(0.0034)
ROA		0.0405***		0.0404***
		(0.0114)		(0.0114)
EquityRatio		-0.2142***		-0.2166***
		(0.0504)		(0.0508)
FedfundsChange		-1.1917***		-1.1930***
		(0.3955)		(0.3955)
Constant	0.4018***	0.8374***	0.4011***	0.8386***
	(0.0065)	(0.0444)	(0.0066)	(0.0445)
Observations	121,671	121,671	121,702	121,702
R-squared	0.7170	0.7397	0.7166	0.7397
Bank_Subsidiary FE	Yes	Yes	Yes	Yes
Quarter FE	No	No	No	No

Table 7: Parent debt and subsidiary bank's risk-taking

This table presents the results of the models described in Section 5.4. The dependent variables are the risk-taking measures (Leverage, LeverageRatio, RiskLeverage, RiskWeight, and Zscore), and the key explanatory variable is parent debt ratio. All models use quarterly data for 1999–2020 and include subsidiary bank fixed effects. T-statistics reported in parentheses are based on robust standard errors clustered at the BHC and quarter levels. Superscripts ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
VARIABLES	Zsore	Zsore	RiskLeverage	RiskLeverage	RiskWeight	RiskWeight	LeverageRatio	LeverageRatio	Leverage	Leverage
ParentDebtRatio	0.1120***	0.0964**	-0.3126*	-0.2847**	-0.0214**	-0.0210**	-0.9791**	-1.2881**	-0.1298	-0.0867
	(0.0405)	(0.0373)	(0.1698)	(0.1377)	(0.0087)	(0.0086)	(0.4375)	(0.5681)	(0.2428)	(0.1726)
Ln AT		0.0988***		-0.3270***		0.0088***		0.5754***		-0.5556***
		(0.0113)		(0.0597)		(0.0032)		(0.2098)		(0.0669)
ROA		-0.0052		-0.4708***		0.0125		-11.5961		-0.9403**
		(0.1023)		(0.1586)		(0.0101)		(9.4560)		(0.3571)
EquityRatio		3.5189***		-14.3605***		0.0816**		-1.2821		-22.9321***
		(0.3634)		(2.0170)		(0.0348)		(1.5466)		(2.8946)
FedFundsChange		-1.9906		-13.0996**		0.0292		-46.0941		-18.8199***
		(1.7717)		(6.1672)		(0.4278)		(52.8379)		(3.0172)
Constant	3.3957***	1.7394***	8.2643***	14.0220***	0.7108***	0.5877***	0.9661***	-6.4359**	11.6670***	21.3019***
	(0.0102)	(0.1538)	(0.0401)	(0.7135)	(0.0023)	(0.0414)	(0.2631)	(2.6420)	(0.0388)	(0.7492)
Observations	121,142	121,142	121,761	121,761	121,763	121,763	104,947	104,947	121,884	121,884
R-squared	0.4477	0.4909	0.6970	0.7408	0.7461	0.7477	0.0713	0.0944	0.6137	0.6995
Bank_Subsidiary FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	No	No	No	No	No	No	No	No	No	No

Table 8: Subsidiary bank's risk-taking and deposit beta estimated with time fixed effects

This table presents the results of the models described in Section 5.4. The dependent variable is the deposit beta estimated with time fixed effects, and the explanatory variables are the risk-taking measures (Leverage, LeverageRatio, RiskLeverage, RiskWeight, and Zscore). All models use quarterly data for 1999–2020 and include subsidiary bank fixed effects. T-statistics reported in parentheses are based on robust standard errors clustered at the subsidiary bank and quarter levels. Superscripts ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

VARIABLES	(1) DepositBeta Time	(2) DepositBeta Time	(3) DepositBeta Time	(4) DepositBeta Time	(5) DepositBeta Time	(6) DepositBeta Time	(7) DepositBeta Time	(8) DepositBeta Time	(9) DepositBeta Time	(10) DepositBeta Time
Zscore	-0.0196	-0.0112***								
	(0.0029)	(0.0021)								
RiskLeverage			0.0102***	0.0066***						
			(0.0011)	(0.0011)						
RiskWeight					0.0425*	0.0666***				
					(0.0230)	(0.0188)				
LeverageRatio							-0.0009	-0.0004		
							(0.0007)	(0.0005)		
Leverage									0.0072***	0.0036***
									(0.0008)	(0.0007)
Ln AT		-0.0335***		-0.0324***		-0.0348***		-0.0385***		-0.0325***
		(0.0032)		(0.0031)		(0.0032)		(0.0034)		(0.0032)
ROA		0.0344***		0.0385***		0.0357***		0.0803		0.0391***
		(0.0060)		(0.0077)		(0.0071)		(0.0817)		(0.0083)
EquityRatio		-0.1927***		-0.1257**		-0.2318***		-0.2181***		-0.1390***
		(0.0453)		(0.0502)		(0.0434)		(0.0465)		(0.0499)
FedFundsChange		-1.9079***		-1.7573***		-1.8507***		-1.8149***		-1.7881***
		(0.2953)		(0.2997)		(0.2990)		(0.3393)		(0.3063)
Constant	0.4229***	0.8509***	0.2723***	0.7373***	0.3264***	0.7874***	0.3539***	0.8824***	0.2720***	0.7525***
	(0.0099)	(0.0427)	(0.0098)	(0.0413)	(0.0166)	(0.0432)	(0.0036)	(0.0448)	(0.0104)	(0.0412)
Observations	125,590	125,590	126,703	126,703	126,705	126,705	106,847	106,847	126,826	126,826
R-squared	0.7093	0.7379	0.7125	0.7373	0.7039	0.7353	0.7404	0.7749	0.7110	0.7356
Bank_Subsidiary FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	No	No	No	No	No	No	No	No	No	No

Table 9: Parent debt and deposit beta estimated without time fixed effects.

This table presents the results from our main analysis in Section 5.5.1. The dependent variable is the deposit beta estimated without time fixed effects, and the key explanatory variable is the parent debt ratio. Model (1) only includes the parent debt ratio as the dependent variable, and Model (2) includes the full set of controls. All models use quarterly data for 1999-2020. T-statistics reported in the paratheses are based on robust standard error clustered at the subsidiary bank and quarter levels. Superscripts ***, **, * indicates statistical significance at 1%, 5%, and 10% levels, respectively.

VARIABLES	(1) DepositBeta	(2) DepositBeta
ParentDebtRatio	-0.0212	-0.0211***
	(0.0000)	(0.0075)
Ln AT		0.0128***
		(0.0033)
ROA		0.0368***
		(0.0053)
EquityRatio		-0.0068
		(0.0403)
Constant	0.3685	0.2019***
	(0.0000)	(0.0442)
Observations	126,828	126,828
R-squared	0.7840	0.7853
Bank_Subsidiary FE	Yes	Yes
Quarter FE	Yes	Yes

Table 10: Relationship between parent debt and deposit beta estimated without time fixed effects - 2SLS analysis.

This table reports the results of the two-stage least squares (2SLS) estimation described in Section 5.5.1. The instrumental variable, MonetaryPolicySurprise (MPS), captures high-frequency changes in interest rates around monetary policy announcements. Model (1) presents the first-stage results, and Model (2) reports the second-stage estimates. All models use quarterly data for 1999–2020. T-statistics reported in parentheses are based on robust standard errors clustered at the subsidiary bank level. Superscripts ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

VARIABLES	(1) Parent DebtRatio	(2) DepositBeta
LagMPS	-0.0016*	
	(0.0009)	
ParentDebt		-0.0116***
		(0.0017)
Ln AT		0.0104***
		(0.0034)
ROA		0.0361***
		(0.0031)
EquityRatio		0.0254
		(0.0396)
Constant	0.0684***	-2.0123***
	(0.0004)	(0.2302)
Observations	121,997	108,367
R-squared	0.0000	0.8186
Macro-level Controls	No	Yes
Bank_Subsidiary FE	No	Yes
Quarter FE	No	No

Table 11: Parent debt and interest expense beta.

This table presents the results from our main analysis in Section 5.5.2. The dependent variable is the interest expense beta estimated with or without time fixed effects, denoted by ExpenseBetaTime or ExpenseBeta, and the key explanatory variable is the parent debt. Model (1) uses ExpenseBetaTime as the dependent variable and includes parent debt as the explanatory variable. Model (2) adds the full set of control variables. Model (3) uses ExpenseBeta as the dependent variable, and Model (4) includes the full set of controls. All models use quarterly data for 1999-2020. T-statistics reported in the paratheses are based on robust standard error clustered at the subsidiary bank and quarter levels. Superscripts ***, **, * indicates statistical significance at 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)
VARIABLES	ExpenseBetaTime	ExpenseBetaTime	ExpenseBeta	ExpenseBeta
ParentDebtRatio	-0.0127*	-0.0123*	-0.0129*	-0.0125*
	(0.0071)	(0.0070)	(0.0072)	(0.0071)
Ln AT		0.0174***		0.0177***
		(0.0034)		(0.0034)
ROA		0.0388***		0.0389***
		(0.0025)		(0.0024)
EquityRatio		-0.0663**		-0.0675**
		(0.0310)		(0.0310)
Constant	0.3758***	0.1550***	0.3875***	0.1629***
	(0.0005)	(0.0455)	(0.0005)	(0.0456)
Observations	126,828	126,828	126,828	126,828
R-squared	0.7718	0.7747	0.7699	0.7730
Bank_Subsidiary FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes

Table 12: Relationship between parent debt and interest expense beta - 2SLS analysis.

This table reports the results of the two-stage least squares (2SLS) estimation described in Section 5.5.2. The instrumental variable, MonetaryPolicySurprise (MPS), captures high-frequency changes in interest rates around monetary policy announcements. Model (1) presents the first-stage results, and Models (2) and (3) report the second-stage estimates. All models use quarterly data for 1999–2020. T-statistics reported in parentheses are based on robust standard errors clustered at the subsidiary bank level. Superscripts ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)
VARIABLES	Parent DebtRatio	ExpenseBetaTime	ExpenseBeta
LagMPS	-0.0016*		
	(0.0009)		
ParentDebtRatio		-0.0090***	-0.0064***
		(0.0018)	(0.0017)
Ln AT		0.0154***	0.0157***
		(0.0035)	(0.0035)
ROA		0.0357***	0.0358***
		(0.0030)	(0.0028)
EquityRatio		-0.0520	-0.0543
		(0.0334)	(0.0331)
Constant	0.0684***	-3.1110***	-2.3255***
	(0.0004)	(0.2316)	(0.2281)
Observations	121,997	108,367	108,367
R-squared	0.0000	0.8089	0.8073
Macro-level Controls	No	Yes	Yes
Bank_Subsidiary FE	No	Yes	Yes
Quarter FE	No	No	No

Figure 1: The quarterly average parent debt ratio

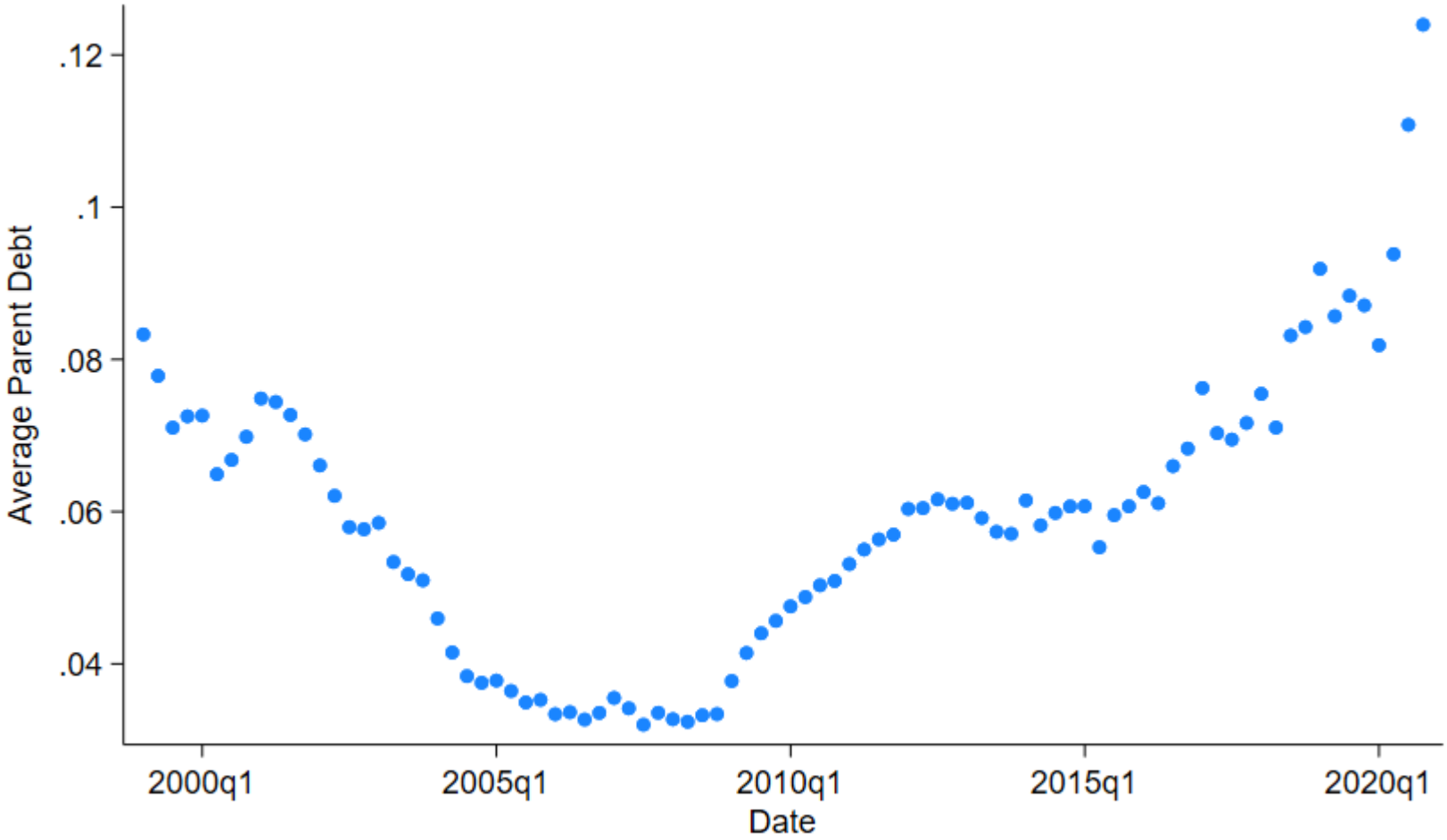


Figure 2: The distribution of deposit beta.

The deposit beta is estimated by regressing the change in a bank's deposit rate on the contemporaneous and previous three quarterly changes in the federal funds rate, using a rolling window of the previous 60 quarters. The deposit beta is winsorized at the 5% level.

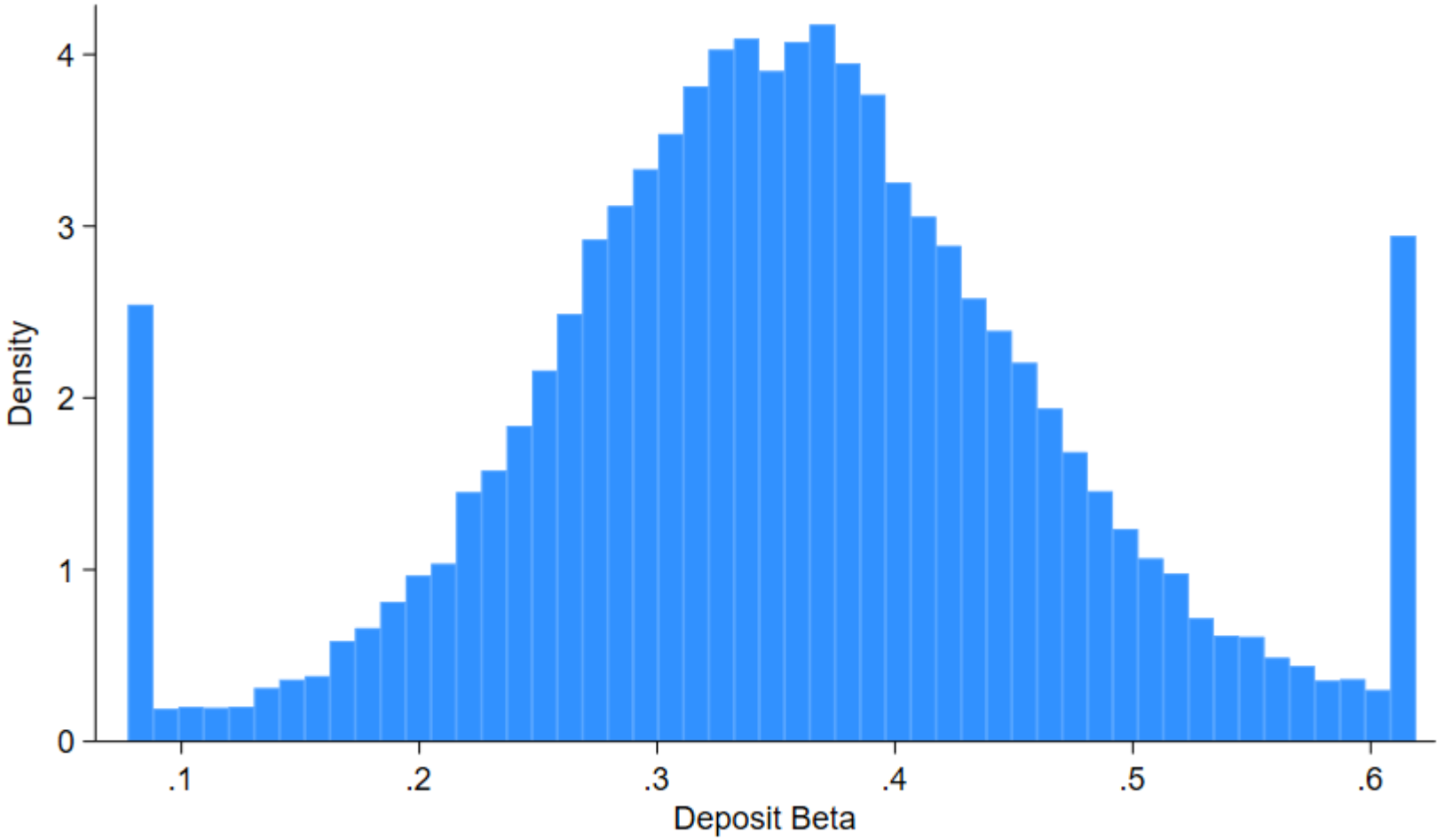


Figure 3: The quarterly average deposit beta.

The deposit beta is estimated by regressing the change in a bank's deposit rate on the contemporaneous and previous three quarterly changes in the federal funds rate, using a rolling window of the previous 60 quarters. For each quarter, we compute the mean deposit beta across all banks. The sample includes banks with at least 60 quarterly observations between 1984 and 2020, and the final analysis sample covers the period from 1999 to 2020. The deposit beta is winsorized at the 5% level.

